

Ionospheric Scintillation Effects on GBAS Ground Multipath Error in Low-latitude Regions

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Introduction

The Ground Based Augmentation System (GBAS) is designed to augment Global Navigation Satellite Systems (GNSS) by providing differential corrections and integrity information to aviation users. GBAS augmentation information corrects the combined effects of multiple sources of errors including satellite clock and ephemeris errors, ionospheric delay errors, and tropospheric delay errors. The residual errors remaining after the correction are calculated as integrity broadcast parameters via theoretical modeling, such as σ_{pr_gnd} in the case of ground system error, to guarantee the integrity of the system using protection level [1][2].

However, when the GNSS signal passes through the irregularities of electron density in the ionosphere, the ionospheric scintillation is accompanied in which perturbation occurs in the amplitude and phase of the GNSS signal. The random propagation paths derived from the ionospheric scintillation intensify the multipath error, and result in larger ground system error than σ_{pr_gnd} according to the existing theoretical model [1][3]. In other words, the integrity of GBAS cannot be guaranteed properly using σ_{pr_gnd} in the case of ionospheric scintillation.

To investigate the effect of ionospheric scintillation on GBAS integrity, this paper assessed the ground system error statistics according to degree of scintillation activity and compared them with σ_{pr_gnd} to evaluate existing GBAS integrity monitoring.

Data

The GNSS data ($L_1 = 1575.42$ MHz and $L_2 = 1227.60$ MHz) was obtained from Low Latitude Ionospheric Sensor Network (LISN), which GNSS stations are located in a region with high ionospheric scintillation activity, around the magnetic equator as described in figure 1 [4].

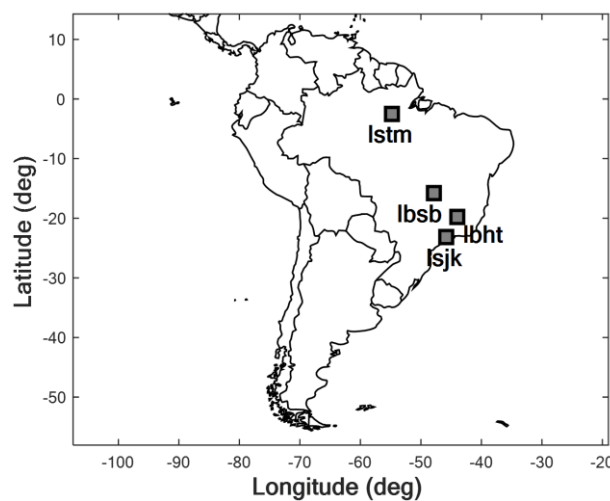


Figure 1. Map of LISN GNSS stations processed in this study

Scintillation days were selected based on amplitude scintillation index, S4. Total 85 days from 2013 to 2014 were selected and input into Long-Term Ionospheric Anomaly Monitor (LTIAM) to estimate the multipath error which corresponds to L1 signal [5][6].

Methodology

A. Scintillation Index Calculation

In this study, the degree of scintillation activity is quantified by amplitude scintillation index, S4. The S4 index is the standard deviation of normalized signal intensity fluctuations as shown in (2). The fluctuation component of the signal intensity is extracted by dividing the input, signal intensity by the detrended output of the low-bandpass filter [7].

$$SI_{fluc} = \frac{SI}{SI_{lbf}} \quad (1)$$

$$S_4 = \sqrt{\frac{\langle SI_{fluc}^2 \rangle - \langle SI_{fluc} \rangle^2}{\langle SI_{fluc} \rangle^2}} \quad (2)$$

The scintillations are classified into 3 main categories as insignificant ($S_4 < 0.1$), Weak & Moderate ($0.1 < S_4 < 0.3$) and Strong ($S_4 > 0.3$) [8]. The assessment is implemented for each category to evaluate the effect according to the scintillation activity on the GNSS signal.

B. Ground System Error Statistic Extraction

The ground system error was computed as multipath error estimates of the GNSS data. The multipath estimates are extracted using linear combination of L1/L2 code and carrier-phase measurements [9]. These linear combinations and resulting multipath error estimates are defined as:

$$MP_{L1}(t_i) = \rho_{L1}(t_i) - \left(1 + \frac{2}{\gamma-1}\right) \varphi_{L1}(t_i) - \frac{2}{\gamma-1} \varphi_{L2}(t_i) \quad (3)$$

$$MP_{L1_new}(t_i) = MP_{L1}(t_i) - Bias_1 \quad (4)$$

In the integrity monitoring of GBAS, integrity parameters are conservatively computed to contain all probability distributions of residual errors. Similarly, in this study, the multipath error statistic is used to cover all probability distributions, conservatively. To extract the statistics, the multipath error is normalized along divided elevation bins and one-minus Cumulative Distribution Function (1-CDF) is obtained. Since actual multipath error distribution has non-Gaussian tails violating the 1-CDF of standard normal distribution. Thus, the one-sigma of normalized multipath error was dilated by multiplying inflation factor (α) to cover the non-Gaussian tails of actual distribution. Lastly, the multipath error statistic was computed as follow.

$$\sigma_{overbound} = |\mu| + f \cdot \sigma \quad (5)$$

Results

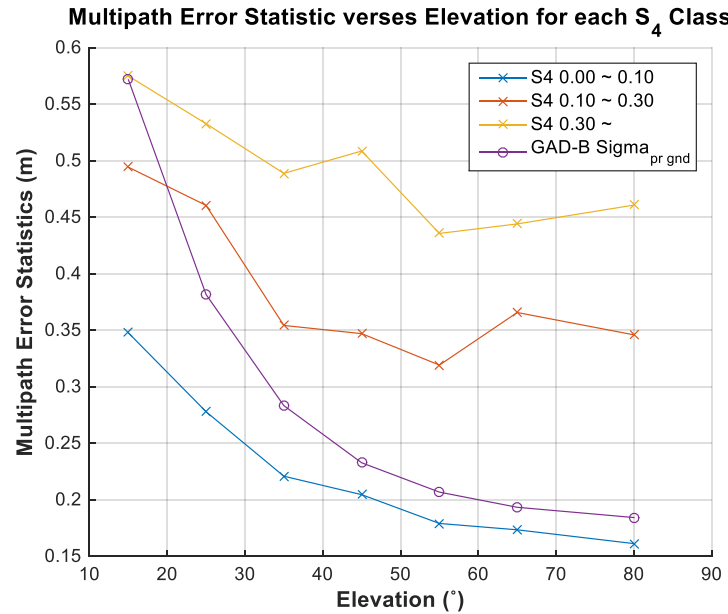


Figure 2. Multipath error statistic verses elevation for each scintillation category

Figure 2 shows the σ_{pr_gnd} derived from theoretical model and multipath error statistic along the elevation bins. The multipath error statistics for insignificant category ($S_4 < 0.1$) are bounded by σ_{pr_gnd} for all elevation bins. However, in the case of weak & moderate category ($0.1 < S_4 < 0.3$), the multipath error statistics violates σ_{pr_gnd} when the elevation bin is above 20. Furthermore, the case of strong category ($S_4 > 0.3$), the multipath error statistics violates for all elevation bins, which means that the theoretical model of σ_{pr_gnd} cannot bound the multipath error in the case of ionospheric scintillation.

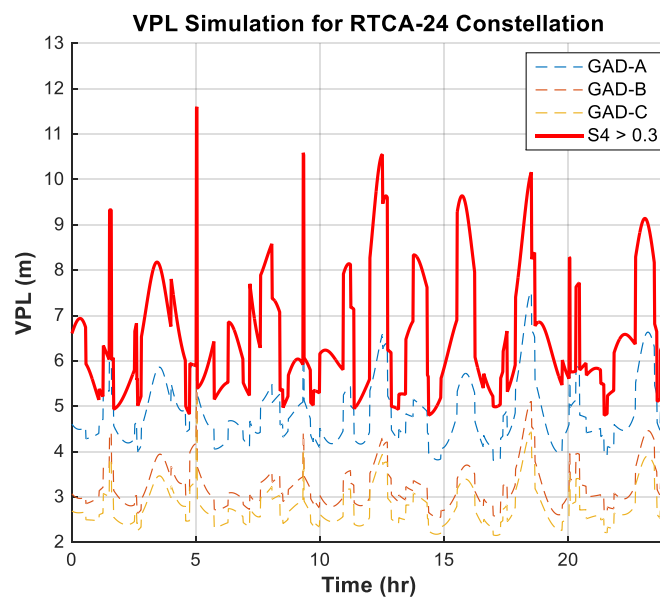


Figure 3. VPL simulation results for multipath error statistics on scintillation case and existing σ_{pr_gnd} model

Figure 3 shows the simulation results of Vertical Protection Level (VPL) to evaluate the effects of scintillation on GBAS integrity. Figure shows that the VPL calculated with multipath error statistic on strong category ($S_4 > 0.3$) exceeds all options of σ_{pr_gnd} model; GAD-A, GAD-B, and GAD-C [10]. It shows that existing σ_{pr_gnd} model cannot guarantee the GBAS integrity in the case of ionospheric scintillation.

Conclusion

In this study, the GBAS error model was assessed by comparing the existing ground system error model and multipath error statistics on scintillation cases. As a result, the existing ground system error model couldn't bound the multipath error statistics on scintillation cases, which shows that the existing GBAS error model cannot guarantee the integrity of the system in the case of ionospheric scintillation.

Based on the results, the graded ground system error model applying scintillation index should be designed to guarantee the GBAS integrity even in the case of scintillation, through further studies.

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